Prodigy Anesthesia Quick Review

Fluid/Blood Replacement

What is the difference between hypovolemia and dehydration?	Hypovolemia describes the loss of extracellular fluid. Dehydration is a disorder of electrolyte concentration in which there is not enough water relative to the sodium level. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 383.*
Why are isotonic crystalloids the fluid of choice for most cases of hypovolemia?	Most cases of hypovolemia (including blood loss) are isotonic losses of extracellular fluid. Thus, the most appropriate replacement solution in most cases is an isotonic crystalloid. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 384.*
What is the intravascular half-life of intravenous crystalloids?	Crystalloid solutions remain in the circulation for about 20-30 minutes. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1165.*
What is the treatment of choice for hyponatremia in a patient with low total body sodium?	In patients with a decreased total body sodium content and hyponatremia, the treatment of choice is 0.9% saline. In patients with hyponatremia and a normal or elevated total body sodium content, the treatment of choice is water restriction. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1117-1118.*
Rank the hypertonic crystalloids D5NS, D51/2NS, 3% Saline, 7.5% NaHCO3, and D5LR from least to greatest according to their tonicity.	D51/2NS (432 mOsm/L), D5LR (525 mOsm/L), D5NS (586 mOsm/L), 3% Saline (1026 mOsm/L), and 7.5% NaHCO3 (1786 mOsm/L) *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1164.*
How will large volumes of normal saline affect pH and electrolyte balance?	Normal saline contains 154 mEq/L of both sodium and chloride. Because the body's bicarbonate concentration decreases as chloride increases, large volumes of saline can produce dilutional hyperchloremic acidosis. *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 366.*
What is the electrolyte composition of Plasmalyte solution?	Na: 140 mEq/L, Cl: 98 mEq/L, K: 5 mEq/L, Mg: 3 mEq/L, Acetate: 27 mEq/L, Gluconate: 23 mEq/L *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1164.*
What is the glucose content of D5W?	50 grams/liter *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1164.*

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Name four crystalloid solutions considered to be isotonic.	Normal saline (308 mOsm/L), D51/4NS (355 mOsm/L), Lactated Ringer's solution (273 mOsm/L), and Plasmalyte (295 mOsm/L) *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1164.*
Name three hypotonic crystalloid solutions	D5W (253 mOsm/L), 1/2 NS (154 mOsm/L), and 1/4 NS (77 mOsm/L) *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1164.*
As a percent of body weight, how much fluid loss would you estimate a patient to have lost if they exhibit an orthostatic increase in heart rate by 15 bpm and decrease in blood pressure by 10 mmHg?	When patients begin to exhibit increases in heart rate by 15 bpm and a decrease in blood pressure by 10 mmHg when moving from lying to sitting, you can estimate that they have lost 15% of their body weight in fluid. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1162.*
What are the pulmonary and radiographic signs of hypervolemia?	Pitting edema, increased urine flow (if renal and cardiac function is normal), tachycardia, pulmonary crackles, pink frothy secretions, wheezing, and cyanosis are all signs of fluid volume overload. Prominent pulmonary vascular and interstitial markings known as Kerly "B" lines are radiographic signs of hypervolemia. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1162-1163.*
What are the laboratory signs of dehydration?	Rising hematocrit (due to concentration of cells in the vascular space), metabolic acidosis, a urine specific gravity greater than 1.010, BUN:creatinine ratio that exceeds 10:1, hypernatremia, urine sodium less than 10 mEq/L, and a urine osmolarity greater than 450 mOsm/Kg. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1162-1163.*
Is cross-matching performed on fresh frozen plasma? Why or why not?	No, fresh frozen plasma (FFP) is evaluated for atypical antibodies and should be the same type as the patient. Cross-matching however, is not necessary. Rh typing is not always matched because an immune response to the Rh antigen has rarely been seen as a result of the transfusion of FFP. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1443.*
What is the incidence of nonhemolytic transfusion reactions and what are the most common symptoms?	Nonhemolytic transfusion reactions occur in about 1-5% of all transfusions. The patient typically exhibits symptoms such as fever, urticaria, and chills. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 398.*

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What are the indications for administration of unmatched blood?	If a patient is exsanguinating and their blood type is known, then an abbreviated crossmatch (that typically takes less than 5 minutes) may be performed. If there is not time to perform any testing, then Type O Rh-negative blood may be administered. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1170.*
What are the signs and symptoms of a hemolytic transfusion reaction?	Fever, chills, nausea, vomiting, diarrhea, hypotension, and tachycardia. The histamine response may result in flushing and bronchospasm. Cytokine release may result in chest and back pain. The patient may also develop hemoglobinuria. As many of these signs are masked during anesthesia, hypotension and signs of microvascular bleeding may be the only symptoms evident. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 426.*
What are the primary reasons that blood products should be warmed when administered in large quantities?	One unit of PRBCs at 4 degrees Celsius will drop the core temperature of a 70 Kg patient by about 0.25 degrees Celsius. Administering large quantities of cold blood products can produce hypothermia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 429.*
How does the administration of stored blood affect the recipient's oxygen delivery mechanism?	The storage of blood is associated with a progressive decrease in adenosine triphosphate (ATP) and 2,3 DPG levels. This results in a leftward shift in the oxyhemoglobin dissociation curve in the red blood cells. Even though the administration of hemoglobin will increase the ability of the blood to transport oxygen, the longer the blood is stored and the more of it that is given, the more the patient's oxyhemoglobin dissociation curve shifts to the left, thereby decreasing the effectiveness of the blood given. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 418.*
How much blood will a full-soaked 4X4 sponge hold?	10 mL *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1167.*
Why are packed red blood cells mixed with a combination of dextrose and adenine (Nutricell or AS-3) or a combination of dextrose, adenine, and mannitol (Adsol or AS-1)?	Packed red blood cells are mixed with about 100 mL of either combination which extends the shelf-life of the blood to 42 days and gives the fluid flow properties similar to that of whole blood. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1439.*

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Which presents the greatest risk of transfusion-related infection, HIV, Hepatitis B, or Hepatitis C?	Hepatitis B carries the greatest risk with an incidence of about 1 in 205,000 per unit administered. Hepatitis C carries an incidence between 1:1,935,000 and 1:3,100,000. No new HUV infections have been attributed to screened blood products since the 1980s. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1448.*
How can infusions of large amounts of banked blood administration affect the patient's ionized calcium level?	Banked blood contains an anticoagulant containing sodium citrate which binds to calcium in the bloodstream. As a result, the administration of large quantities of banked blood can result in sharp decreases in the patient's ionized calcium level. Normally, however, a healthy, normothermic patient can metabolize the amount of citrate found in 20 units of blood per hour. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 397.*
What immunoglobulin is primarily responsible for the antibodies associated with a transfusion reaction?	The antibodies produced against a blood type that does not match the recipient's are primarily IgM antibodies. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1169.*
With regards to blood product administration, what is leukocyte reduction and why would it be performed?	Leukoreduction is the reduction of white blood cells in transfused blood through the use of filters. It is performed to reduce the incidence of nonhemolytic transfusion reactions. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 398-399.*
How much of an increase in hemoglobin and hematocrit would you expect to see after administering a unit of packed RBCs?	In general, each unit of PRBCs elevates the hematocrit by 2-3% and the hemoglobin by 1 mg/dL. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 396.*
What blood antibodies do patients with Type A blood naturally possess? Type B? Type AB? Type O?	Persons with Type A blood possess anti-B antibodies. Conversely, patients with Type B blood possess anti-A antibodies. Persons with Type AB blood do not possess antibodies to either blood type and persons with Type O blood possess both anti-A and anti-B antibodies. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1169.*

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Are blood products screened in the laboratory for West Nile Virus? Malaria?	It is tested for West Nile virus, but not malaria. Rapidly developed testing procedures have reduced the risk of transmission of West Nile virus by transfusion. Donated blood is not currently tested for malaria. Instead, donors are screened during the interview process for relevant symptoms or travel to areas where the diseases are more prevalent. Malaria is relatively uncommon in the United States. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 399.*
What is CPDA and why is it added to blood products? What is a disadvantage conferred by the addition of CPDA?	CPDA (citrate-phosphate-dextrose-adenine) is added to whole blood when collected to act as an anticoagulant and provide the blood with a shelf-life of about 35 days. Within 24 hours of its addition, the platelets are dysfunctional and many coagulation factors are below normal levels. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1438.*
What is the point of administering washed red blood cells?	Washed red blood cells are rinsed with isotonic saline solutions. This removes plasma proteins, leukocytes, and platelets. Doing so helps prevent severe allergic reactions mediated by recipient IgE antibodies and donor plasma proteins. It also helps prevent reactions caused by pre-existent antibodies to IgA in the donor plasma which can result in anaphylaxis in IgA-deficient recipients. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1439.*
What is acute normovolemic hemodilution?	Acute normovolemic hemodilution is an alternative to transfusion in which whole blood is removed from the patient prior to surgery with replacement of the lost volume in the form of crystalloids. By diluting the vascular bed, any blood lost during surgery will have a lower hematocrit. The previously removed blood is then reinfused back to the patient when intraoperative blood loss has ceased. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 400.*
What is the most common cause of bleeding following massive transfusion?	The most common cause of bleeding after massive transfusion is dilutional thrombocytopenia. Coagulation and platelet studies should guide the administration of FFP and platelets during massive transfusions. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1175.*
What is the incidence of bacterial contamination of platelets and packed red blood cells?	Bacterial contamination is the second leading cause of death related to transfusion. The incidence of a positive culture from a specimen derived from a bag of platelets is 1:2000 and the incidence is 1:7000 from packed red blood cells. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 702.*

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Why is the incidence of bacterial contamination higher with platelets than with red blood cells?	Platelets are stored at 22-24 degrees Celsius, which is more supportive of bacterial growth than the refridgeration required of packed red blood cells. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 424.*
What is the difference between blood typing and blood crossmatching?	Blood typing (a type and screen) determines the patient's blood type and predicts compatible transfusions 99.94% of the time. Crossmatching, which is the actual mixing of the donor and recipient blood in a 'trial transfusion' increases the possibility of a compatible transfusion by only one-hundredth of 1%. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 395.*
At what serum platelet levels would an infusion of concentrated platelets be indicated?	Transfusions of platelet concentrates are usually indicated when the platelet concentration is less than 50,000/microliter, and are certainly not indicated at a platelet concentration higher than 100,000/microliter. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 397.*
How long can platelets be stored at room temperature? What is the overall risk of bacterial contamination via platelet transfusion?	Platelets can be stored for a maximum of 5 days at room temperature and have a risk of bacterial contamination of 1 in 12,000. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 399.*
What coagulation factors are present in fresh frozen plasma and what are its general indications for administration?	All coagulation factors are present in fresh frozen plasma. It may be administered for reversal of the effects of warfarin, bleeding due to dilutional coagulopathy, or treatment of microvascular bleeding due to prolonged prothrombin time or partial thromboplastin time. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 397.*
What is the recommended dose of fresh frozen plasma for the reversal of warfarin? What is the dose in all other disorders?	The dose of FFP for the reversal of warfarin is 5-8 mL/kg. For all other purposes, the dose should be 10-20 mL/kg. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 397.*
What are the components of cryoprecipitate?	Cryoprecipitate contains factor VIII, factor XIII, von Willebrand factor, and fibrinogen. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 398.*
What are the most common indications for the administration of cryoprecipitate?	Cryoprecipitate is most commonly administered in the treatment of patients with von Willebrand disease or patients with low fibrinogen levels (less than 80 mg/dL). *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 398.*

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How does the concentration of fibrinogen found in a single unit of cryoprecipitate compare to that found in a unit of fresh frozen plasma?	The normal adult dose of cryoprecipitate contains between 100 and 250 mg of fibrinogen, or about four times the amount found in a unit of fresh frozen plasma. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1443.*
Is it necessary to match cryoprecipitate to the recipient's blood type?	Cryoprecipitate can contain anti-A or anti-B antibodies and should therefore, be typed to the patient's blood type. The incidence of nonhemolytic reaction (fever, chills, etc) is similar to that of red blood cell transfusions. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1444.*
What is the electrolyte composition of Normosol-R?	(all values in mEq/L) Na: 140, K: 5, Mg: 3, acetate: 27, gluconate: 23 *Miller RD, Pardo MC. Basics of Anesthesia. 6th ed. Philadelphia: Elsevier Saunders; 2011: 366.*

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